

DEEPAK KUMAR PRASAD

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📍 Bangalore, India (Open to relocation)

SUMMARY

Software Architect with 9+ years designing large-scale supply chain planning platforms serving 150+ global clients. Deep expertise in distributed systems, solver integrations, multi-layer configuration frameworks, and performance engineering—delivering 50% compute reduction and 30% memory optimization in production environments. Proven ability to define technical direction, lead 10+ person engineering teams, and translate complex domain problems into scalable platform architecture.

EXPERIENCE

o9 Solutions, Bangalore

Senior Application Architect

Application Architect

Senior Software Engineer

Python Developer (Contract)

Feb 2019 – Present

Apr 2025 – Present

Apr 2021 – Apr 2025

Aug 2019 – Apr 2021

Feb 2019 – Aug 2019

- **Architected the Supply Planning platform configuration framework**—a multi-layer system (Product Default, Product Override, Project Config, Tenant Config) supporting namespace-partitioned overrides, hot-reload via atomic replacement, and schema-validated merge precedence. Eliminated config drift across 150+ client deployments and enabled zero-downtime tenant onboarding.
- **Re-engineered core Supply Planning compute pipeline** (TLB, MRP, IOP) with advanced parameterization and memory-efficient data structures, achieving **50% reduction in batch computation time** and **30% decrease in memory usage** across multiple production environments.
- **Designed and integrated 6+ third-party optimization solvers** with the o9 platform—including Samsung NSR solver using Boost serialization and Parquet-based data exchange—standardizing the solver integration contract and improving deployment scalability across client environments.
- **Designed foundational Demand Netting and Order Promising plugins**—now mission-critical modules in production for 150+ global clients—with extensible configuration hooks and observability instrumentation for enterprise-scale deployments.
- **Built an internal developer productivity platform** (Tenant Extractor, Builder, Performance Analyzer, Test Recorder/Runner, Rule Creator, Dataset Multiplier) adopted by 100+ internal teams, improving consultant and developer productivity by **25%** and reducing environment setup time significantly.
- **Established a modular plugin versioning system** managing 30+ plugin variants, enabling independent deployment and customization per client without disrupting the core platform—reducing release risk and accelerating client onboarding cycles.
- **Defined technical roadmap and led engineering team of 10+ developers:** conducted architecture reviews, established coding standards, drove performance benchmarking culture, and mentored engineers on supply planning domain and distributed systems design.

Primesoft Solutions Inc, Bangalore

Associate Engineer

Jun 2016 – Jul 2018

- **Architected a workflow orchestration microservice** on Netflix Conductor, customizing the open-source framework for enterprise orchestration requirements and integrating it into production infrastructure.
- **Modernized a large-scale enterprise security platform** by integrating 5+ Google APIs and refactoring core subsystems, improving system reliability and reducing technical debt across the codebase.

- **Built Attendance & Visitor Management systems** (Android) automating check-ins and visitor tracking, reducing manual overhead by **70%** with actionable facility analytics.

EDUCATION

Bachelor of Engineering, Computer Science
SJCE, Mysore

Aug 2012 – May 2016

SKILLS

Technical Skills

Languages	Python, C++, Java, SQL
Architecture	Distributed Systems, Microservices, Multi-layer Config Frameworks, API Design, Performance Optimization, Solver Integration
Frameworks & Tools	FastAPI, Boost (C++), ReactJS, Streamlit, LangGraph
Data & Messaging	PostgreSQL, Parquet, ETL Pipelines, Data Serialization
AI / Automation	LLM Orchestration (LangGraph, multi-agent patterns), Claude API, Python automation tooling
DevOps	Docker, Kubernetes, Git, CI/CD, Linux
Domain	Supply Chain Planning (MRP, TLB, IOP, Demand Netting, Order Promising), Technical Leadership

CERTIFICATIONS

Design for Performance Architect — o9 Solutions · Credly	Oct 2023
Claude AI Development (Specialization, 3 courses) — Anthropic Education	Apr 2026
Python for Everybody (Specialization, 5 courses) — University of Michigan · Coursera	May 2017
Introduction to Data Science (Specialization, 4 courses) — IBM · Coursera	Jan 2021
Programming Foundations with JavaScript, HTML and CSS — Duke University · Coursera	Dec 2017

Full portfolio: beingdpkpr.github.io/deepak-prasad/